

AI for All

We Need to Talk, AI: a comic essay, carried into the lecture

Prof. Dr.-Ing. Mark Schutera | September 14, 2026 · 7 parts

Seven parts, one comic

Part	Section	The question it answers
I	Intro	why talk about AI at all
II	Basics	what AI is, and what made it work
III	Examples	where it already runs
IV	Chances	what it is good for
V	Risks	what it costs us
VI	Outlook	who gets to shape it
VII	Social Utopias	what we would want it for

cover, the section spiral

How to read this deck

weneedtotalk.ai



One comic page, one slide, **all six panel captions**. Nothing on the following slides is my wording. The frame title is the page's own heading, the six captions are lifted verbatim and numbered in the comic's own reading order, and the page number under them tells you where to read the rest.

The comic is a free read at weneedtotalk.ai under CC BY-NC-SA 4.0. It is an essay, not a textbook: it argues. Argue back.

page 5, the authors at the table

[1]

1

Intro

pages 6 to 10

Knowing what future we would like to have

“

Knowing the future is impossible. But what we can do:
Knowing what future we would like to have. And then
work on it.

We Need to Talk, AI | page 6

page 6, the opening spread

Our creatures have always turned against us

1 Our creatures have always turned against us. This applies to the Golem of Jewish mythology as well as to Frankenstein.

3 Parents whose kids prefer to wear rubber boots instead of weather-proof sandals even when it's hot outside know that.

5 In 2011, Stephen Hawking said that 'Success in creating AI would be the biggest event in history.'

2 Neither magic nor natural sciences have ever helped to make artificial humans submissive and controllable.

4 An AI Take-Over is a hypothetical scenario in which AI takes control of the planet – taking it away from us.

6 'Unfortunately, it might also be the last, unless we learn how to avoid the risks.' How can we make AI systems both safe and beneficial?

We Need to Talk, AI | page 7

page 7, Golem, Frankenstein and the rubber boots

Who is Julia

1 Good question. That's something I ask myself, now and then.

3 I love data's friendliness, their lack of strategic answering. If they are biased it's not their fault but ours.

5 Artificial Intelligence (AI) might have the potential to help us find answers to problems we are not able to solve.

We Need to Talk, AI | page 8

2 I love making sense of our world. I love to discover patterns in data, with code.

4 Our brains are prone to some errors. Overconfidence. And discrimination against people who seem to be different from us.

6 For that reason, we need to demystify AI. Therefore, I'm writing this comic. Let your voice be heard.

page 8, Julia, the digit wall and the 404

Who is Lena

1 Not one thing in particular, I guess.

3 Visuality is one of the most vivid encryption systems I know. We are constantly increasing and renewing it collectively.

5 AI is a field that gives us an intimate view on society's desires and fears.

2 I love visual language. I love to encrypt thoughts into pictures by exploiting society's visual memory.

4 It's fun to mess around with visual codes by recomposing associations. Pictures start to shift meanings and interfere in society's processes.

6 Looking from different angles might help us to overcome commonly known perspectives. Let your view be seen!

We Need to Talk, AI | page 9

page 9, Lena, the iceberg of visual codes

Lessons from AI in Fiction

1

3

5

2

4

6 Seems, like we should not seek to replicate but to augment ourselves.

We Need to Talk, AI | page 10

page 10, the five fictional robots, 1927 to 2014

2

Basics

pages 11 to 19

What is Artificial Intelligence?

1 Actually, the methods are not new. Just the techniques and the data.

3 If you define intelligence as the ability to capture the essence of a new situation...

5 This being said, AI is still in its early stages (although it reads differently on the news).

2 At the moment, AI is mainly concerned with performing complicated but repetitive tasks.

4 ... real AI needs to think more like us. With intentionality, with emotions, with conscience; capturing counterfactuals.

6 Up to now, it's more adequate to speak of 'artificially augmented intelligence'.*

We Need to Talk, AI | page 13

page 13, the essence of a new situation

Computer Power

- 1 What kind of techniques helped establish AI? One key indicator is computer power.
- 2 Moore's law is the observation that the power of computers will double every 18 months.
- 3 The amount of computer power used for training AI has gone up to 3,5 month-doubling time.
- 4 The growth has largely been driven by computer hardware, optimized to train and run AI-models.
- 5 This explains why AI has been growing faster and faster...
- 6 ... together with advances in data availability and algorithms (=the recipes) behind AI.

We Need to Talk, AI | page 14

page 14, the cost and computer-power curve

Algorithms 1

1 A second key indicator are better algorithms. Now we can teach machines instead of programming them.

3 The more complicated the task, the harder it was to automate.

5 Nowadays, AI learns through trial and error, based on data we feed it. The more data, the more accurate it becomes.

2 Before, when intending to automate a task using AI, we had 1) to understand it completely and 2) to program it.

4 The breakthrough was in understanding how we learn to perform tasks.

6 Modern AI applications achieve breathtaking results that we could hardly imagine a few years ago.

We Need to Talk, AI | page 15

page 15, a cat? a cat? a dog..?

Algorithms 2

1 What do these buzzwords even mean? Machine Learning; Deep Learning; Neural Networks (NN)...?

3 (Note: The AI bookshelf includes other topics, as well.)

5 DL uses artificial Neural Networks (NN) that learn patterns directly from the data they are fed.

2 Machine Learning (ML) helps recognizing patterns based on existing data and algorithms: an important branch of AI.

4 Before Deep Learning (DL), we had to know a lot about the data we were programming. Otherwise, ML would break down.

6 Like our brain, NNs compare new information with objects they know. However, they have not yet been able to think of new objects.

page 16, the AI bookshelf

We Need to Talk, AI | page 16

Data 1

1 Data is the third element that makes AI so powerful. But what exactly is data?

3 Data describes what we know and we can draw conclusions from it. Data can take the form of text, numbers, images, or sounds.

5 The more data an algorithm is trained on, the better its results: conclusions, predictions, timing, actions.

2 Any form of raw fact or figure is data. Whether on paper or in electronic form.

4 The internet and mobile devices like smartphones, drones or simple sensors have made data abundant and far more valuable.

6 Better results = higher usability = more users = more power. Therefore 'data is the new oil'.

We Need to Talk, AI | page 17

page 17, the data word cloud

Data 2

1 Okay, all AI applications need training data to get good results.

3 Too much data for us to process. Just the right amount for AI.

5 Four out of five necessary steps have to do with data.

2 Only with the invention of the internet, we were able to collect and store enough data to make AI work.

4 There is an adaption of Maslow's Pyramid of Needs. Aimed at achieving AI solutions.

6 AI is a journey that begins with data: collecting, storing, transforming and exploring it.

We Need to Talk, AI | page 18

page 18, the data pyramid, four steps of five

General AI

1 In the 1960s, AI pioneers hoped that machines could soon learn to think without human intervention.

3 Until the 2000s, AI was a dirty word that 'simply didn't work'. But when data, hardware and infrastructure were ready in 2010, a wide range of AI applications followed.

5 No AI can represent causal relationships or integrate abstract knowledge, e.g., what objects are, what they are for, and how they are typically used.

2 But from 1973 onwards, disappointment and criticism in the community, followed by pessimism in the press, led to the 1st 'AI Winter'. A 2nd one followed.

4 Despite their impressive progress and success, today's AI is narrow. Its tasks are often classification and need a lot of data and a lot of energy.

6 We really need to adjust our expectations and stop over-hyping AI. If we don't, we may find ourselves in another AI Winter.

We Need to Talk, AI | page 19

page 19, the two AI Winters

3

Examples

pages 20 to 27

Maps

1 Using location data from smartphones...

3 ... in real-time.

5 The proprietary algorithms of Maps have access to vast amounts of data.

We Need to Talk, AI | page 21

2 ... the AI-application Maps can analyze how fast traffic is moving...

4 Additionally, maps incorporate user-reported traffic incidents.

6 This means, Maps can suggest the best and fastest routes for us.

page 21, two phones, the delta and the route

Recommendation

1 All major internet companies – or ‘platforms’ – refine their products and processes using AI.

3 Likewise, streaming providers use AI to analyze our consumer habits down to the last detail.

5 Also, the platforms use the knowledge of our habits to create and offer new products similar to previous successes...

2 If we’re looking for books or shoes, internet shops use AI so that our purchase decision gets easier.

4 We benefit from improved suggestions and rediscoveries of old pearls which we have missed so far.

6 ...or to let AI autonomously stir those components together that have worked great elsewhere.

page 22, the component tag cloud

We Need to Talk, AI | page 22

Social Networks

- 1 Those of us who use social networks use AI frequently, without necessarily knowing it.
- 2 Social networks use AI to recommend friends, news, photos, tags, advertisements.
- 3 They even 'curate' the posts of your friends according to general or personal interests – with the help of AI.
- 4 Now imagine a network that knows even more than our habits, relations, thoughts, interests, likes and dislikes, our hobbies and whereabouts.
- 5 Our state of health or bank account for example.
- 6 In China, this already exists in the form of 'WeChat'. Just recently, the Chinese government officially announced to supervise each group with 12+ members.

We Need to Talk, AI | page 23

page 23, what a wonderful match

Self-driving Cars

1 Today, we can already activate the autopilot function which automatically and comfortably drives our car or flies our plane.

3 Within a few years, this might change.

5 Or maybe not. Nearly every car accident involves some sort of unforeseen circumstances, the AI must confront for the first time.

We Need to Talk, AI | page 24

2 But ultimately, we are responsible. The systems are not autonomous – or self-driving.

4 We would have plenty of room to sleep, work, read or play, sober or drunk after a party.

6 Moreover, user tests show that we need autonomous systems to explain their decision to feel safe. Not easy.

page 24, the unforeseen circumstance

Identifying Images

1 A lot of AI work deals with identifying patterns. At an enormous scale.

3 Since 2018, everybody can use a free tool helping AI beginners detect images.

5 ...sometimes it is life-threatening. The military is a huge fan of it, too.

2 In 2015, computers made fewer mistakes than we do sorting images into predefined categories.

4 While early image identification could be lifesaving...

6 Unfortunately, AI is susceptible to hallucinations: caused by biased data or by hackers who, for example, make AI mistake guns for apes.

We Need to Talk, AI | page 25

page 25, rifle 2017 next to Picasso 1910

Emotional AI

- 1 Only ~ 10% of the emotional meaning of a message is conveyed through words.
- 2 To really understand us, you need our facial expressions, our tone of voice, our gestures plus our words.
- 3 As our relationship with AI is becoming more and more intimate...
- 4 ... we need our AI devices (like assistant systems, cars, or robots) to sense and adapt to what we mean and not only to our words.
- 5 Indeed, Emotional AI could help us with some highly valuable tasks. Help autistic children, for example, to learn emotions.
- 6 Nonetheless, our freedom of thought might be at stake in hostile scenarios.

We Need to Talk, AI | page 26

page 26, ten per cent of the message

Exoskeletons

1 Exoskeletons are an external framework we can wear to augment our natural physical ability and reduce strain and weaknesses.

3 We need assistance that is specifically fit for us. We need to tailor our devices at an individual level.

5 This way, the AI can get our individual profile right. AI can make our exoskeletons fit our needs – elderly man or firefighter.

We Need to Talk, AI | page 27

2 But every one of us is different.

4 AI can use real-time measurements of our body signals, our breathing rate or hip extension for example.

6 Our bodies are different and constantly changing. Nowadays, the only feasible way of translating this to robotics is through the help of AI.

page 27, elderly man or firefighter

4

Chances

pages 28 to 36

Dealing with Big Data

1 Our power to analyze the tons of data we are now able to store and to collect can be a bottleneck.

3 Even with hundreds of us analyzing patterns in big data, the sheer volume simply overwhelms our capacities.

5 Traditional algorithms are bound to follow the same logic over and over, without flexibility or learning new paths with new data.

2 AI helps us deal with large and complex datasets in ways we have never seen before.

4 Even if we could analyze data, but it proves to be exhausting and tedious, we can use rule-based AI systems – expert systems – that do the job for us.

6 Now enter AI. Because AI systems get smarter as more data is given to them, they are well suited for patterns and anomalies over the time.

We Need to Talk, AI | page 29

page 29, hundreds of us against the volume

Efficiency

1 In many situations, AI is simply faster and more accurate than we are. Let's look at an example*.

3 Competitors had to review 5 legal contracts in 4 hours and identify 30 legal issues.

5 Humans achieved, on average, an 85% accuracy rate, the AI 95%. Humans took 92 minutes to complete the task, the AI... 26 seconds.

We Need to Talk, AI | page 30

2 A legal platform carried out a competition between 20 experienced lawyers against a trained AI.

4 They were rated according to how accurately they identified each issue. Who won?

6 The human lawyers who competed against the AI in the experiment said, the tasks were very similar to what lawyers do every day.

page 30, twenty lawyers against one AI

Cognitive Biases

1 Our brain, as a result of evolution, was not optimized for rational decision-making or perfect diagnoses.

3 Our brain power is biologically limited; therefore, we work with heuristics, trust our gut feeling or store & retrieve wrong evidence.

5 And this is what makes our distortions have large-scale consequences. Were they random, they would cancel out.

We Need to Talk, AI | page 31

2 Instead, we strive for a competitive degree of 'fitness' in our specific environments; quite fast and often satisfying.

4 We call this cognitive bias. There are many of them and we share them to some degree.

6 Used wisely, AI could indeed help us improve our diagnoses and decisions.

Creativity

1 AI can save time otherwise wasted on stupid, tedious work for creativity. But it can do even more.

3 When an AI puts together a playlist of up-and-coming producers of house music, it may inspire our next DJ-set.

5 Judging is an ability we possess. And we love doing it. This is where AI together with us can achieve great things, with a human tweak.

We Need to Talk, AI | page 32

2 When an AI sorts our photos, it is easier to make a photo-calendar with our children (in 28 positions) for our parents as a present.

4 We can also get creative directly with the AI, compiling data and coding it, in a narrow area. But the AI has no judgement.

6 The smaller the role of humans in the process of art-creation becomes, the more AI questions traditional concepts of art.

page 32, just a generic style transfer

Connection

1 Most doctors became doctors because they wanted to connect with people in order to heal them.

3 And it's still only retrospective, error-prone knowledge.

5 AI can support proactive, individualized strategies.

2 Today, they often act like overpaid bookkeepers: taking in and spitting back data, prescribing drugs, adjusting doses, ordering tests.

4 But we cannot connect to doctors who struggle to keep up with messy data from disparate connections. And they cannot connect to us.

6 Our doctors could get to know us better, having the time to learn how a disease uniquely affects us and educate us in the right way.

We Need to Talk, AI | page 33

page 33, the doctor as overpaid bookkeeper

New Opportunities

1 AI allows for fascinating new opportunities. Teaching robots without a single line of coding, for example.

3 Every time a task or location changes, the robot's software must be re-programmed. This is time-consuming, expensive and difficult.

5 AI now enables us to teach robots how to move, either together with the robot, or, with jackets and gloves containing sensors.

2 Robots solve complex tasks today – but these tasks are deterministic. At any point of time, you know in advance where your robot needs to be.

4 Even for experienced programmers, it is hard to come up with a set of verbal rules of how to move when movements and settings are tricky or unpredictable.

6 From the recorded example-movements, automation scripts can be created and optimized – without us having to write complex code. Fantastic!

page 34, the sensor jacket and glove

We Need to Talk, AI | page 34

Inclusion

1 AI is also about helping us, about improving inclusion and our well-being, by removing barriers.

3 AI helps us type (with words) or talks to us. Users of all ages are adopting virtual assistants because it is so simple to learn how to handle them.

5 In the classroom, AI can identify individual challenges and offer more personalized approaches.

2 AI reads to us, it narrates the world around us and help us 'see'. Speech-to-text AI helps us 'hear', too.

4 Smart home devices help us do things easily around the house and live independently.

6 AI virtual nurses or therapists offer 24-hour-support, or someone to talk to in case we are shy.

We Need to Talk, AI | page 35

page 35, a wall missing several bricks

Comfort

1 AI satisfies and reinforces our desire for comfort.

3 It is tempting not to have to leave our couch at all, no matter what we want to do: eat, work, chat, watch a movie, listen to music...

5 All institutions with predominantly analogue offerings are undergoing a tremendous process of change. But it might be worth it.

We Need to Talk, AI | page 36

2 Almost all answers to our consumer, communication or information needs are just a mouse click or an app away. And most of the time, AI is involved.

4 'Pre-AI'-institutions such as shops, local transport, restaurants, libraries, or doctors must react to our new convenience.

6 It is precisely these institutions that are social places where we can come together and exchange ideas. Where we can decelerate. 'Waste' time.

page 36, mall_blueprint.pdf

5

Risks

pages 37 to 43

Surveillance

1 AI makes monitoring people much easier. Not only via our words or facial expressions, via our consumer behavior or interests.

3 Weak radio waves can penetrate walls, but our bodies reflect them. Using this data, an AI can be trained to identify our movements.

5 That means the AI does not only know where we are. The AI can also distinguish what we are doing, whether we are standing, or eating.

2 But also, via our movement patterns. Each one of us moves differently. Our movement patterns are unique.

4 The AI can translate our movement patterns to little stick figures. Once registered, the AI can even identify us behind a wall.

6 It really comes close to magic. Hello, Harry Potter! And hello, surveillance state.

We Need to Talk, AI | page 39

page 39, the stick figures behind the wall

Opinion-Forming and Media

1 AI influences public opinion-forming and the media. Our public sphere has become fragmented.

3 New players and sources make special-interest-content popular and visible and, thus, important for mass media.

5 Social media-users receive AI-personalized content of political organizations, based on users' personality profiles.

2 Borders between private and public communication are blurry. User-generated content is rich but polarized and often privatized on closed platforms.

4 Right-wing activists, for example, use AI bots, too, to spread hate and fake news in social media and, ultimately, mass media.

6 Could we, in turn, use AI to enable a better opinion-forming – and identify those who benefit from current AI tools?*

We Need to Talk, AI | page 40

page 40, the fragmented public sphere

Future Work

1 The share of jobs at high risk of replacement by computers or robots has increased significantly.*

3 ... as well as creative jobs like artists, chefs or journalists, or blue-collar jobs such as construction workers or drivers.

5 We are better in understanding non-trivial, new situations, building relationships, creating context and meaning. Custom-fit solutions.

We Need to Talk, AI | page 41

2 This involves all kinds of jobs on the social ladder. White-collar jobs like lawyers, medics, or brokers...

4 AI can outperform us in doing repetitive tasks. Cheap, strong, fast, easy.

6 We are also needed as partners of AI: developing algorithms that work, supervising, repairing. Combining us with AI.

page 41, the job ladder

Data Security and Safety

- 1 Our blind trust in AI providers is misplaced. We all know that the amount of data that we are implicitly or explicitly forced to leave behind is too great.
- 2 Since AI needs tons of data, we need to improve our data security and safety. A broad campaign, starting with, e.g.:
- 3 The 'data letter', an obligation for all companies and authorities to tell us what data they store about us and how to view, correct and delete it;**
- 4 Penalties for data outflows for system operators, and an obligation for connected devices to declare the frequency of the security updates (incentives);
- 5 Permanent government investment in secure IT, e.g., open source components written and updated in secure programming languages;
- 6 And: a widely available education in safe programming, along with educational, advisory and complaint services for the safe use of our digital systems.

We Need to Talk, AI | page 42

page 42, the data letter inbox

Energy Use

- 1 In fact, AI has huge potential to optimize energy use, by analyzing complex data, making forecasts, optimizing processes via brain-like algorithms.
- 2 But conventional hardware was not designed for AI, hence it needs much more computing power, much more energy, than our brains do.
- 3 An AI supported translation service, e.g., DeepL, can handle 5,1 quadrillion operations per seconds. But that means masses of computing power and data.
- 4 Today, data centers use more energy than some countries, e.g., Iran.* In 2018, we had 33 Zettabyte digital data per year. By 2025, we'll have 175.
- 5 One of the most worrying forecasts says data centers will use 8% of global energy in 2030. The big internet firms bet on renewable energies.
- 6 In short, our brains are far more energy efficient than our best computers – and we still do not really understand how.

We Need to Talk, AI | page 43

page 43, save energy, save the planet

6

Outlook

pages 44 to 49

Overall Human-Machine Intelligence

- 1 We mainly interact with AI via our smartphone. A proprietary marketplace for countless uncured applications (=platform economy).
- 2 On the smartphone, we use AI basically for entertainment (image recognition), communication (language and translation) and information (e.g. google).
- 3 Unbeknownst to us, AI improves performance and battery life, too. And: It analyzes us, its users.
- 4 Up to date, there is no smartphone that is especially good at helping us get better ideas or interpret information. There is no interface for that either.
- 5 The development budgets are driven by numbers of purchase. In turn, our purchase decision is based on entertainment and communication values, and the design of the device.
- 6 But couldn't we also develop a smartphone that makes us more creative, more intelligent? That understands like us? Really augment us?

We Need to Talk, AI | page 45

page 45, the smartphone as marketplace

Feminism

1 The contribution of women to AI is usually underestimated, especially concerning the democratic applicability of computing.

3 What we fear is not an AI-takeover. We fear scientists' homogeneity leading them to (finally!) 'create' intelligent beings – in an egocentric way.

5 And many of those whose service jobs will be replaced belong to an already vulnerable group: minority women.

We Need to Talk, AI | page 46

2 They were the first computers, actually: running calculations in the first information network of the world.

4 In an AI world, where 'female' service robots are 'hot' and submissive, how can children learn to treat women with respect, with dignity?

6 We need (different!) women to participate in the development and use of AI. Otherwise we'll just reproduce patriarchal structures.

page 46, Ada Lovelace and Grace Hopper

Digital Capitalism

- 1 Algorithms have always been an integral part of our modern capitalist societies.
- 2 The idea of objectiveness, general abstract rules, quantification, de-personalization and the concept of equality are characteristics of modernity.
- 3 Max Weber, e.g., pictures the judge as an automation who processes concrete cases (data) based on an algorithm, namely the law.
- 4 Marx' workers are an appendage of the machine, alienated from the spiritual potencies of the capitalist working process.
- 5 Digital capitalism is the ground on which AI thrives. The central paradigm is – instead of production and sale – the collection and use of data.
- 6 Since today's AI is largely incomprehensible to us, we need to regulate and shape AI as a public service and snatch the omnipotence over AI from the few dominant, digital companies.

We Need to Talk, AI | page 47

page 47, Weber's judge and Marx's worker

ART: Accountability, Responsibility, Transparency

1 Throughout the world, we share a growing awareness: We need to make AI safe, beneficial and fair for us.

3 Accountability is the concept that AI should be held responsible for the results of its algorithms.

5 Transparency refers to the need to describe, inspect and reproduce the AI algorithms and results, and to manage the data used, in a fair way.

We Need to Talk, AI | page 48

2 This requires the participation and commitment from many of us. It means training, regulation and awareness in 3 equal pillars:

4 This includes our responsibility to develop a framework for AI that incorporates our values.

6 But let's not fool ourselves. For this ART of AI, we need a new and more ambitious form of governance.

page 48, the three equal pillars

No Fear of Numbers

1 In order to understand how AI could improve our life, we propose a real revolution:

3 Many of our brains signal 'pain' when we read numbers, think or talk about them.

5 Zero is also part of the binary code, fundamental to today's computers and their language. Fundamental to AI, therefore.

We Need to Talk, AI | page 49

2 Healing the public shaming in our individual math-biographies that caused fear.

4 Not only does this deprive life. Zero, e.g., is a fascinating tool allowing us to write '1908' instead of MCMVIII: $1,000 + (1,000 - 100) + 5 + 1 + 1 + 1$.

6 With improved 'math-esteem', more of us could and should discuss chances and risks of AI in our lives.

page 49, 1908 against MCMVIII

7

Social Utopias

pages 50 to 53

AI today is not Skynet

“

For over 300,000 years we've been hunting with spears;
for less than ten years we've been looking for our next
song with the help of AI.

We Need to Talk, AI | page 51

page 50, the section opener

It depends on the training data

“

If we train AIs on discriminatory data, they will make discriminatory decisions and reduce, not increase, the opportunities, and resources of the underprivileged.

We Need to Talk, AI | page 52

page 52, the application procedure

Raising the question, not settling it

“

An AI may not be able to resolve all injustices, but it can raise questions about how we want to live.

We Need to Talk, AI | page 52

page 52, the motorway through the city

Or a creepy surveillance state

“

Or is this vision making us create a creepy surveillance state? Why should the authorities always be well-meaning? How could we make sure they are?

We Need to Talk, AI | page 53

page 53, the same platform, read twice

What do you think?

“

But the first step has been made: What can AI help us with? Where do we have to be careful? What do you think? Let your voice be heard and your point of view be seen.

We Need to Talk, AI | page 53

page 53, Tbc.

Further reading, the book's own

Authors	Title
Agrawal, Gans, Goldfarb	Prediction Machines: The Simple Economics of Artificial Intelligence, 2018
Evans	Broad Band: The Untold Story of the Women Who Made the Internet, 2018
Chollet, Allaire	Deep Learning with R, 2018
Lee	AI Superpowers: China, Silicon Valley, and the New World Order, 2018
Bostrom	Superintelligence: Paths, Dangers, Strategies, 2014
Zuboff	The Age of Surveillance Capitalism, 2018
Russell, Norvig	Artificial Intelligence: A Modern Approach, 2010
Daum	Das Kapital sind wir: Zur Kritik der digitalen Ökonomie, 2017

page 55, the reading list

[1]

If you want to build one

the track



This deck is about what AI does to us. For how it is built, work Google's **foundational courses** track, four courses in order:

Course	Length	Answers
Introduction to Machine Learning	20 min	what ML is
Machine Learning Crash Course	12 modules	how to build one
Problem Framing	45 min	whether to at all
Managing ML Projects	90 min	how to keep it alive

developers.google.com/machine-learning/foundational-courses

a laptop on the same table

[2]

References

- 1 Julia Schneider and Lena Kadriye Ziyal. *We Need to Talk, AI: A Comic Essay on Artificial Intelligence*. Berlin: Dr. Julia Schneider, 2019. ISBN: 978-3-748531-28-9. Free read, CC BY-NC-SA 4.0. <https://weneedtotalk.ai>.
- 2 Google Developers. *Foundational Courses: Machine Learning*. Course hub. <https://developers.google.com/machine-learning/foundational-courses>. 2025.
- 3 Google Developers. *Introduction to Machine Learning*. <https://developers.google.com/machine-learning/intro-to-ml>. 2025.
- 4 Google Developers. *Machine Learning Crash Course*. <https://developers.google.com/machine-learning/crash-course>. 2025.
- 5 Google Developers. *Introduction to Machine Learning Problem Framing*. <https://developers.google.com/machine-learning/problem-framing>. 2025.
- 6 Google Developers. *Managing Machine Learning Projects*. <https://developers.google.com/machine-learning/managing-ml-projects>. 2025.

**Let your voice be heard
and your point of view be seen**

Prof. Dr.-Ing. Mark Schutera | DHBW Ravensburg